

SPECK User Guide

Substrate Point Enumeration and Classification Kit

This guide walks through a typical SPECK session, from building an organism code list through exporting your point-count data. Numbered callouts on each screenshot point to the item described in the text next to it.

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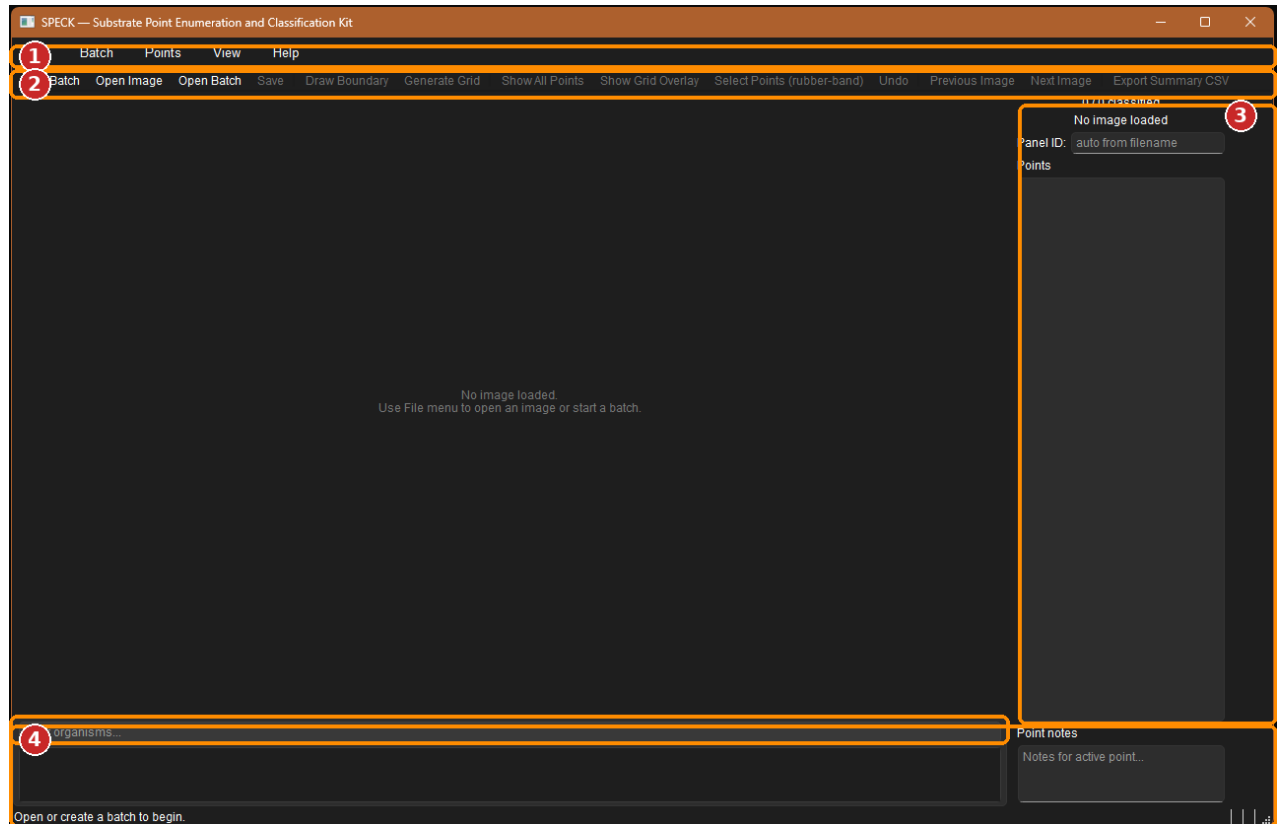
Repository: github.com/bransond-sms/SPECK

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1. What SPECK does

SPECK replaces manual point-count analysis (such as CPCe) with a modern desktop tool. For a batch of images, you draw a boundary, lay down a grid of points, click through the grid assigning an organism code to each point, and export the results as CSV files for analysis. A batch covers a whole directory of images and can be saved and resumed across sessions.

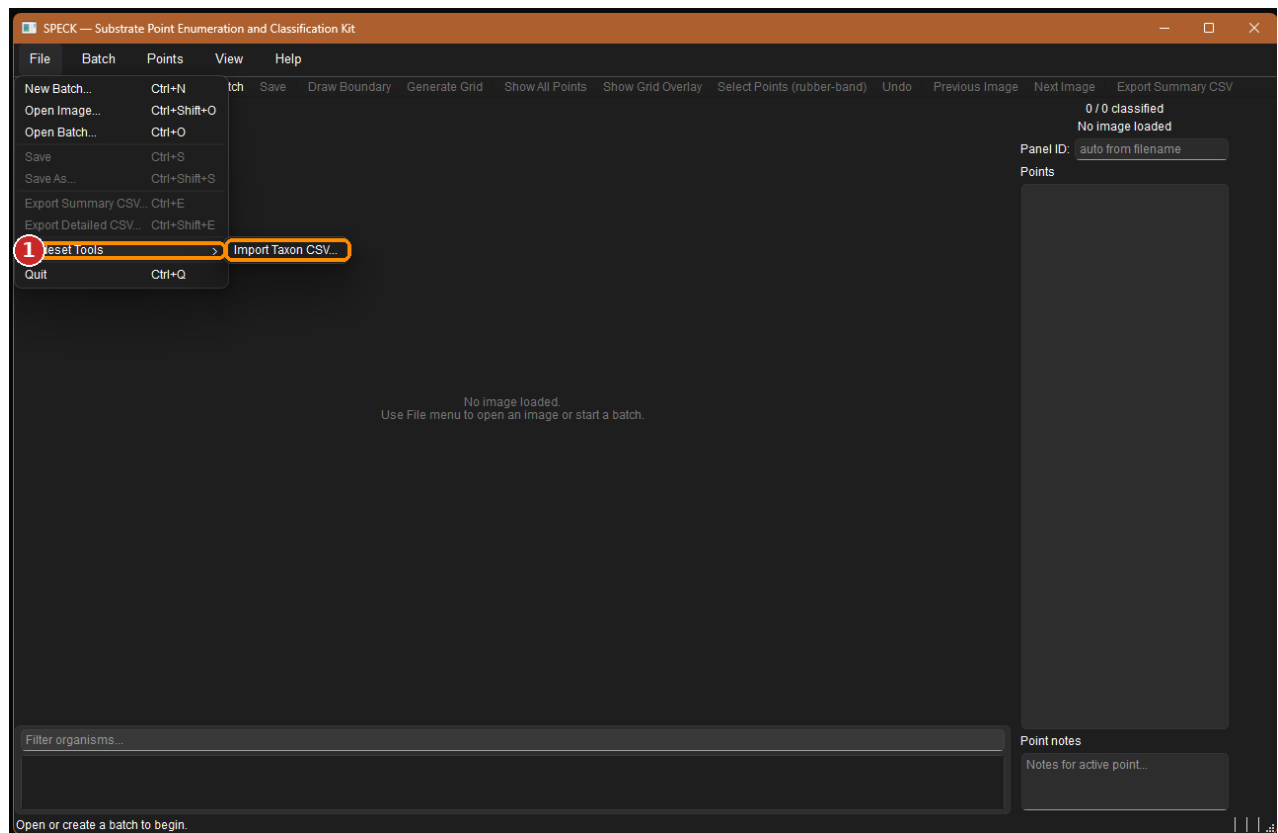


1 Menu bar 2 Toolbar (main actions) 3 Points panel and current image status 4 Organism code filter and code buttons

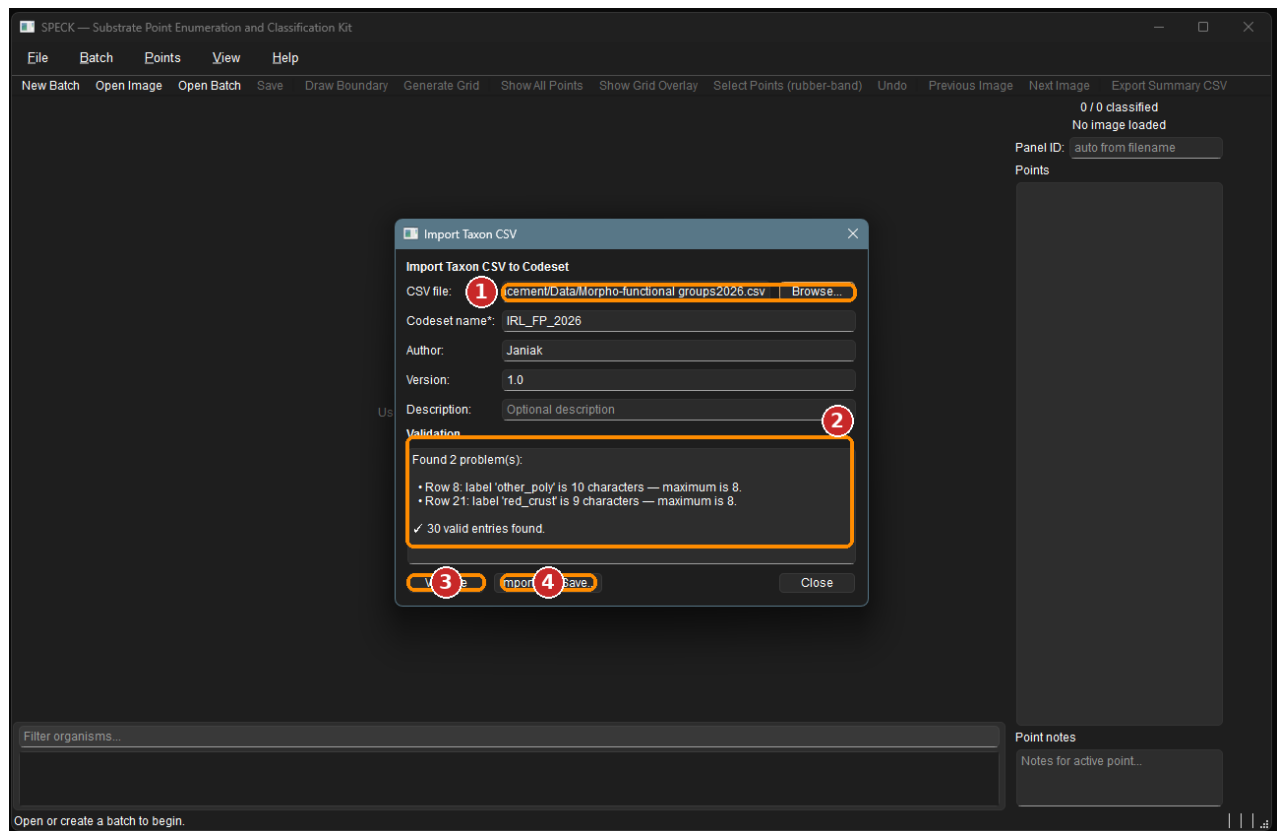
2. Build an organism code list (codeset)

Before starting a batch, SPECK needs a codeset: the list of organism or substrate codes you will classify points into. The easiest way to create one is to import it from a CSV file, for example an existing species or morpho-functional group list.

Open File → Codeset Tools → Import Taxon CSV.



In the dialog that opens, browse to your CSV file and give the codeset a name, author, and version. Click Validate to check the file — SPECK will flag any problems, such as codes longer than the 8-character limit, before you import. Once the file is valid, click Import and Save to create the codeset (a portable JSON file you can reuse or share).

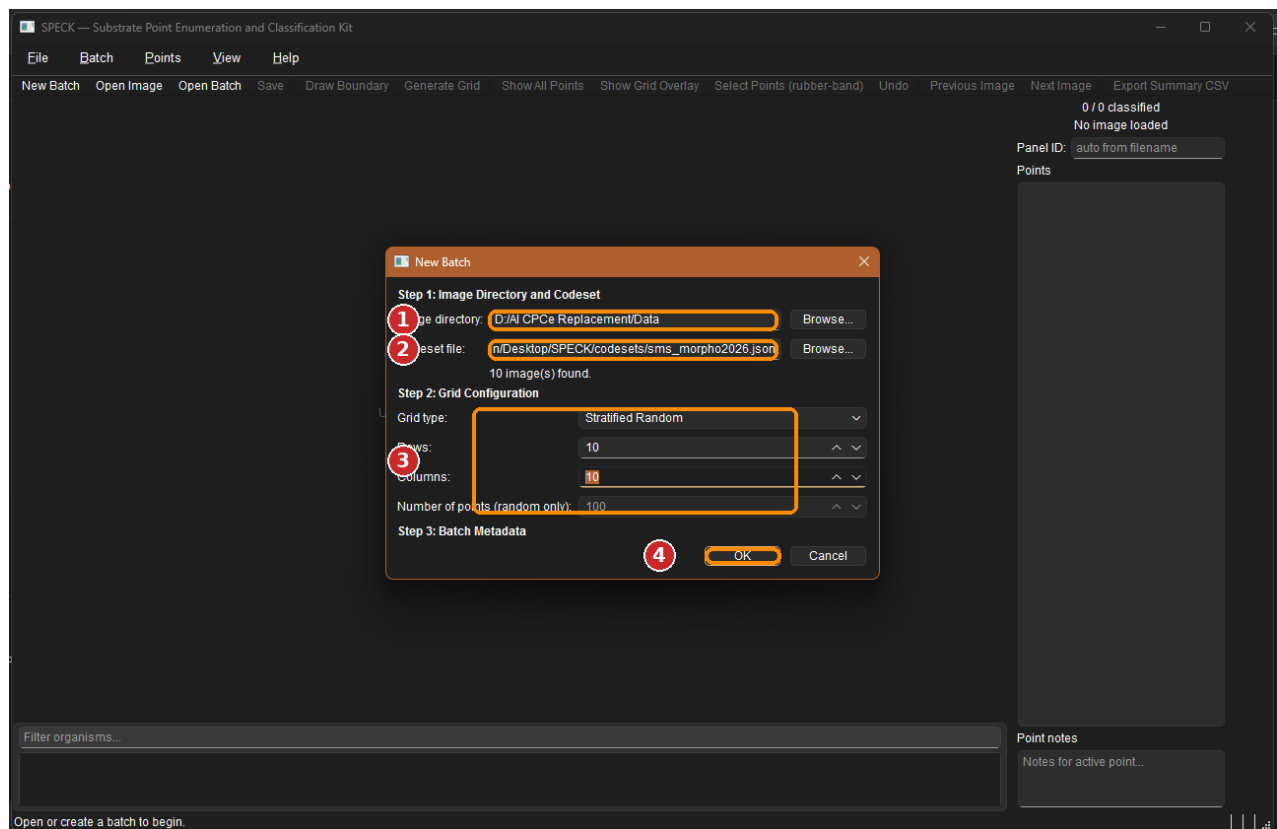


1 Browse to the CSV file 2 Validation results (fix flagged rows in the CSV and re-validate if needed) 3 Validate 4 Import and Save

Colors for each code are assigned automatically by SPECK. There is no manual color-picking step — every code in the list gets a distinct color as soon as the codeset is created.

3. Start a new batch

Use File → New Batch to begin working on a directory of images. This single dialog brings together the three things a batch needs.

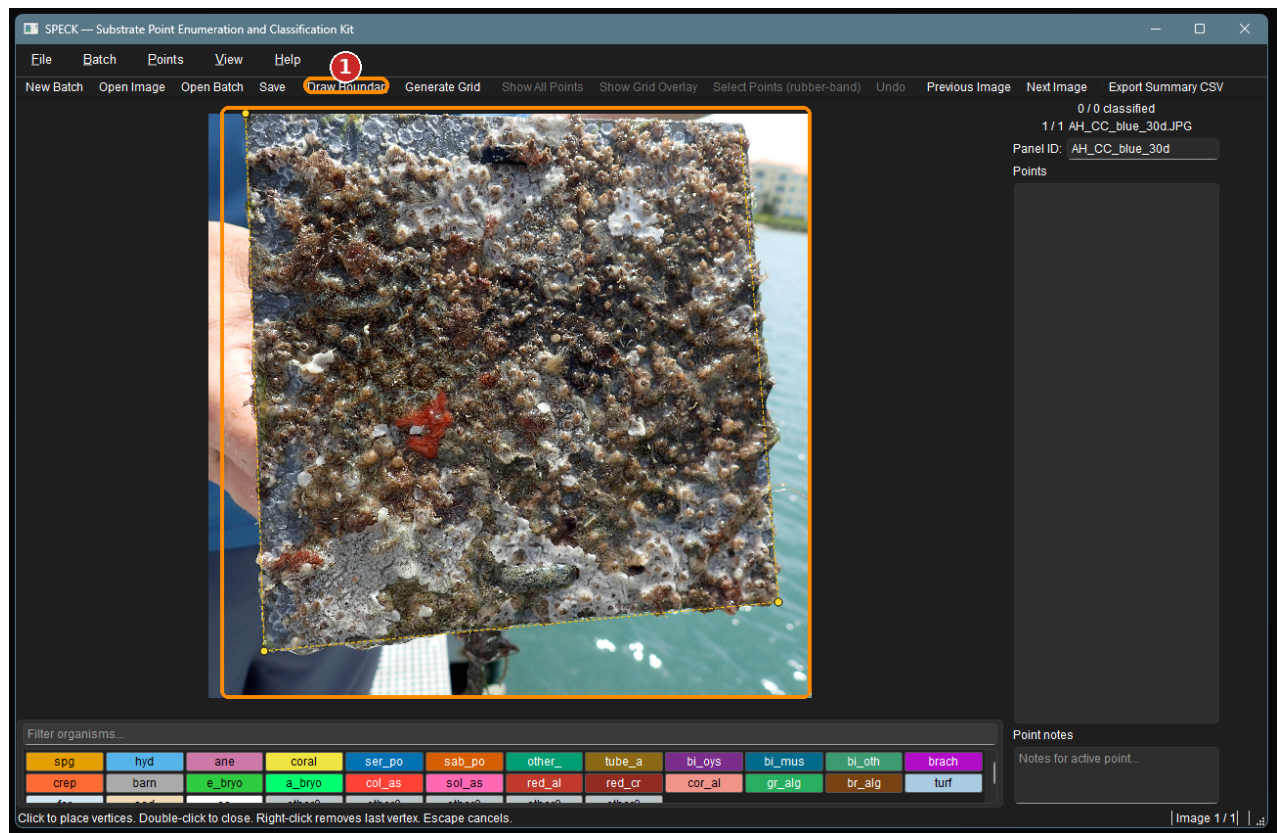


1 Image directory — the folder of images to analyze 2 Codeset file — the JSON codeset created in the step above 3 Grid configuration — grid type, rows/columns or number of random points 4 OK to create the batch (a Batch Metadata step below also captures site, dates, and analyst)

A batch is saved as a single .speck file that covers the whole image directory, so you can close SPECK and resume the batch later from File → Open Batch.

4. Draw the boundary on an image

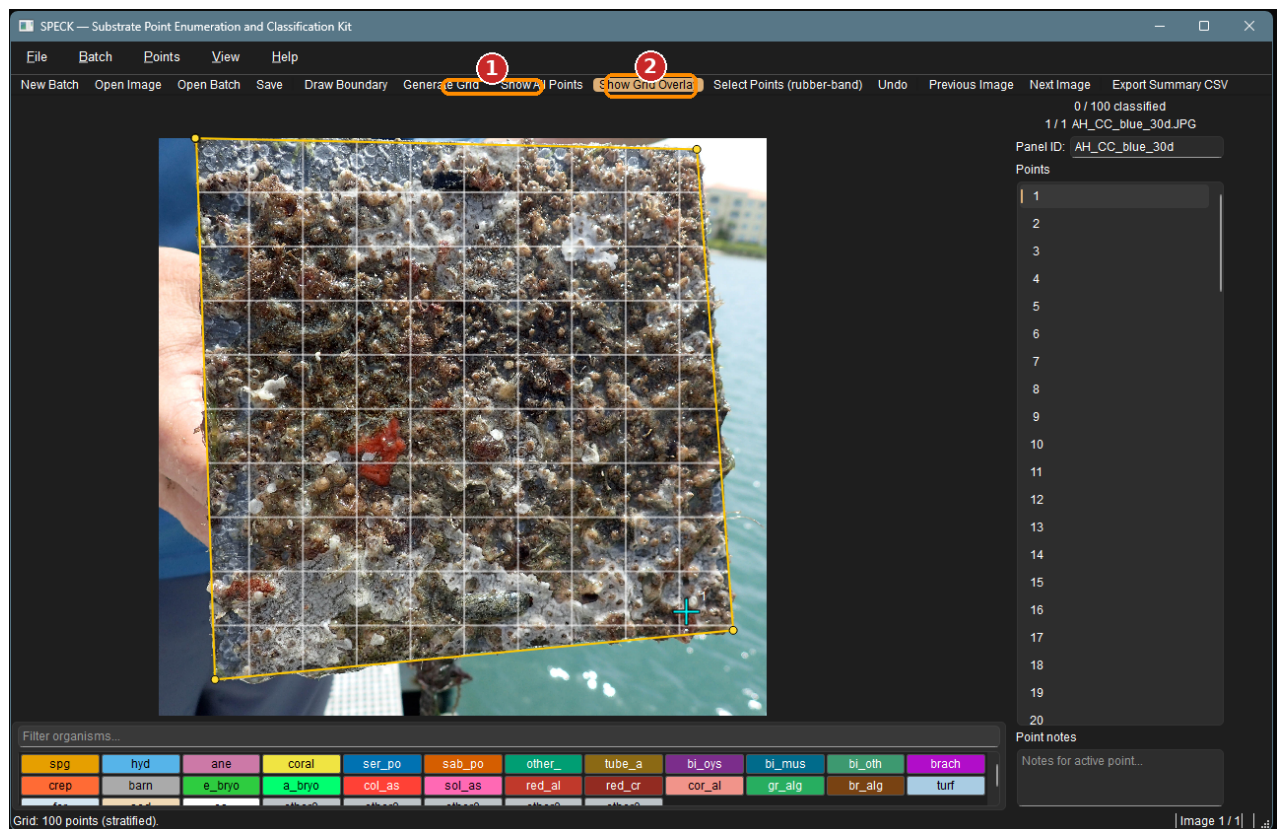
With an image open, click Draw Boundary in the toolbar, then click to place vertices around the area you want to sample — for example, the edge of a settlement panel. Double-click to close the boundary; right-click removes the last vertex; Escape cancels.



1 Draw Boundary toolbar button

5. Generate the point grid

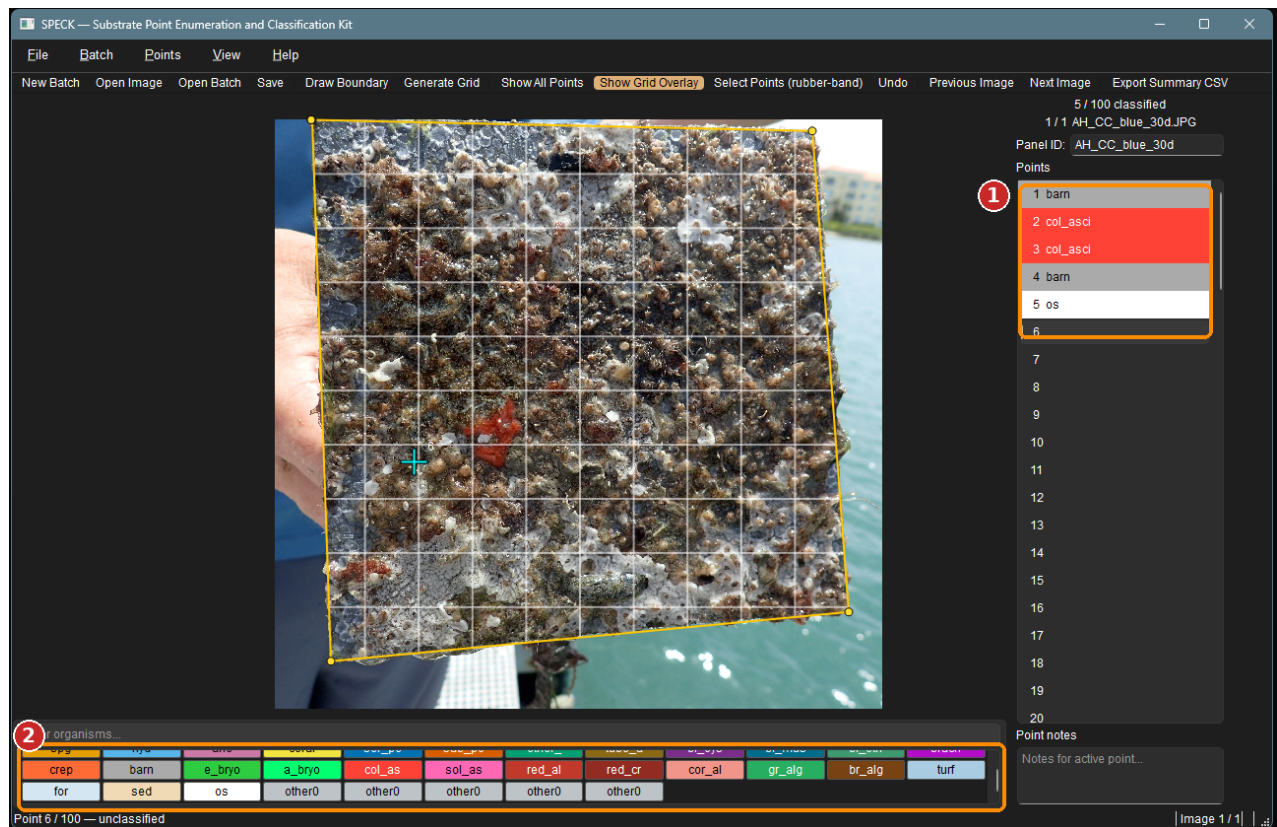
Once a boundary is drawn, click Generate Grid to lay down points inside it, using the grid settings chosen when the batch was created (or adjust them again from the Generate Grid dialog). Show Grid Overlay toggles the grid lines on and off over the image.



1 Generate Grid 2 Show Grid Overlay (shown here toggled on)

6. Classify points

Click a point in the image (or select the next one in the Points list) to make it active, then click the matching code button along the bottom to classify it. Classified points are listed with their code next to the point number; the Filter organisms box narrows the code buttons by name when a codeset is long. A point notes field is available for the active point if you need to record something specific about it.



1 Points list — classified points show their code (unclassified points show just the number) 2 Organism code buttons, filterable by the box above them

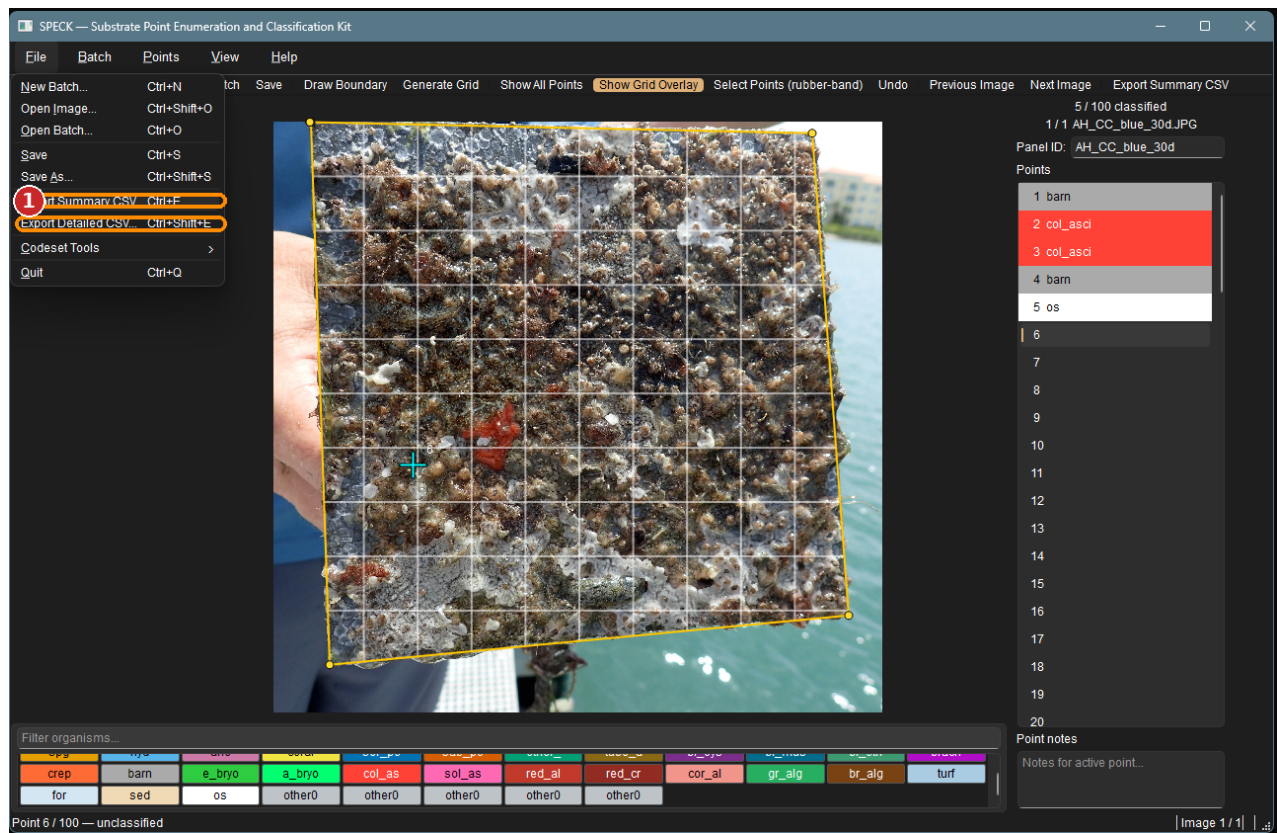
The classified count in the top right (for example “5 / 100 classified”) tracks progress through the current image. Use Next Image / Previous Image in the toolbar to move through the batch.

7. Save your work

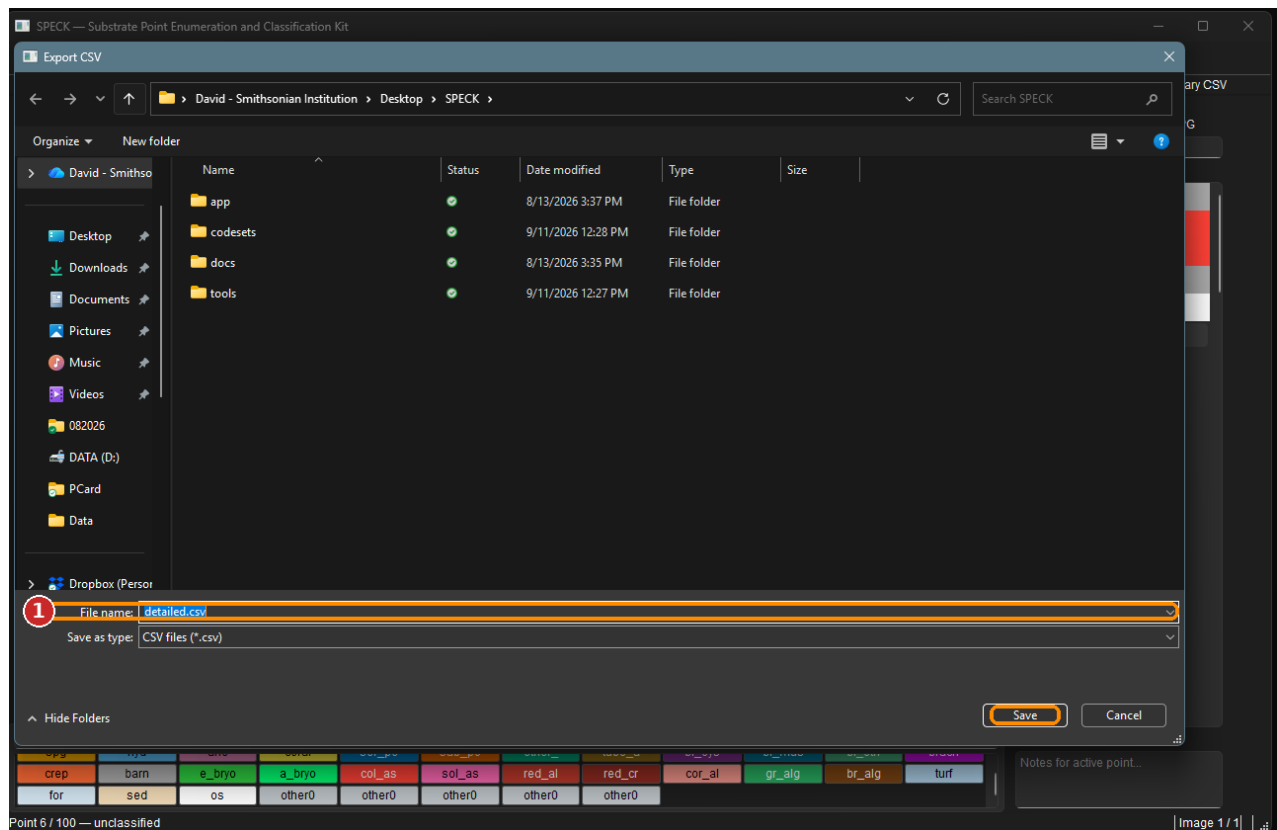
Save (or Save As) in the File menu or toolbar writes your progress back to the .speck batch file, including boundaries, grids, and classifications completed so far. Because a batch covers the whole image directory, saving regularly lets you stop partway through a large batch and pick up again later from the same point.

8. Export results

When you are ready to analyze your data, use File → Export Summary CSV or Export Detailed CSV. Summary export gives per-image totals by code; Detailed export gives one row per point, which is usually what you want for further analysis.



1 Export Summary CSV and Export Detailed CSV



1 Choose a location and file name, then Save

The exported CSV is in long format: one row per point per image, with batch-level metadata (site, panel ID, photo file, GPS) alongside each point's coordinates, assigned organism code, and any point notes.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	site_code	location	panel_id	deployme	deployme	deployme	retrieval_y	retrieval_r	retrieval_c	data_colle	sample_p	photo_file	gps_lat	gps_lon	gps_alt	point_nun	point_x	point_y	organism	point_notes		
2			AH_CC_blue_30d									AH_CC_blue_30d.JPG				1	3094.83	2778.5	Barnacles			
3			AH_CC_blue_30d									AH_CC_blue_30d.JPG				2	2267.72	2834.49	Colonial ascidians			
4			AH_CC_blue_30d									AH_CC_blue_30d.JPG				3	3251.53	2541.11	Colonial ascidians			
5			AH_CC_blue_30d									AH_CC_blue_30d.JPG				4	287.81	1474.34	Barnacles			
6			AH_CC_blue_30d									AH_CC_blue_30d.JPG				5	694.82	51.17	Open Space			
7			AH_CC_blue_30d									AH_CC_blue_30d.JPG				6	816.75	2006.94				
8			AH_CC_blue_30d									AH_CC_blue_30d.JPG				7	1149.84	1906.23				
9			AH_CC_blue_30d									AH_CC_blue_30d.JPG				8	1572.04	452.92				
10			AH_CC_blue_30d									AH_CC_blue_30d.JPG				9	1795.23	2936.3				

A detailed CSV export opened in a spreadsheet program

9. Quick reference

Task	Where to find it
Create a codeset from a species list	File → Codeset Tools → Import Taxon CSV
Start a batch (image folder + codeset + grid)	File → New Batch
Open an existing batch	File → Open Batch
Draw a sampling boundary	Draw Boundary (toolbar)
Lay down / show the point grid	Generate Grid, Show Grid Overlay (toolbar)
Classify the active point	Click a code button along the bottom
Move between images in the batch	Previous Image / Next Image (toolbar)
Save progress	Save / Save As (File menu or toolbar)
Export per-image totals	File → Export Summary CSV
Export point-level data	File → Export Detailed CSV

Panel ID is filled in automatically from the image filename but can be edited per image. Colors for organism codes are assigned automatically when a codeset is created or imported and are not user-selectable.